

LAO CAI, Viet Nam

WMO: 488030

Lat: 22.4969N

Lon: 103.9660E

Elev: 97

StdP: 100.17

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	11.3	6.9	6.2	13.4	8.3	6.9	13.1	6.7	17.6	5.7	18.7	0.3	140	0.268

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.7	36.0	27.2	34.9	27.1	34.0	26.9	28.5	33.0	28.1	32.6	27.7	32.0	2.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.4	23.6	30.7	27.0	23.0	30.2	26.6	22.5	29.9	93.3	33.2	91.2	32.8	89.5	32.3	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.5	4.4	3.6	8.1	38.3	1.6	1.2	7.0	39.2	6.0	39.8	5.1	40.5	4.0	41.3
			7.6	29.5	1.5	0.9	6.5	30.2	5.6	30.7	4.8	31.2	3.7	31.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	17.3	19.1	22.0	25.6	28.0	29.3	29.2	28.9	27.8	25.5	22.1	18.4
	DBStd	5.05	3.28	3.97	3.40	3.01	2.56	1.90	1.75	1.71	1.73	2.11	2.61	2.69
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	144	62	37	8	0	0	0	0	0	0	0	3	33
	CDD10.0	5274	226	253	371	467	558	580	597	586	536	480	362	259
	CDD18.3	2376	28	57	121	217	299	330	338	327	286	222	116	34
	CDH23.3	23249	64	262	641	1861	3315	4072	4084	3767	2953	1633	512	84
(13)	CDH26.7	9264	3	55	162	667	1462	1879	1794	1589	1111	444	93	5
(14)	Wind	WSAvg	0.9	0.9	1.1	1.2	1.3	1.1	0.9	0.8	0.8	0.7	0.7	0.7
(15) Precipitation	PrecAvg	2042	42	46	80	151	249	301	400	379	181	105	64	44
	PrecMax	2514	109	126	231	272	477	505	538	642	367	193	159	186
	PrecMin	1607	0	12	6	30	53	175	245	127	78	0	1	3
	PrecStd	232	29	30	52	50	111	77	69	135	65	59	45	44
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	30.2	32.2	35.5	37.6	37.5	37.0	36.2	34.7	32.9	30.6	27.3
		MCWB	20.7	21.9	23.4	25.8	26.8	27.7	27.2	27.4	26.7	24.7	23.7	21.4
	2%	DB	25.2	28.5	30.1	33.2	35.4	35.6	35.3	34.9	33.6	31.5	29.1	25.6
		MCWB	19.8	21.7	23.4	25.9	27.1	27.4	27.4	27.1	26.2	24.7	22.9	20.4
	5%	DB	23.4	26.6	28.5	31.7	33.9	34.3	34.2	33.9	32.7	30.5	27.7	24.0
		MCWB	19.1	21.1	22.9	25.5	26.7	27.3	27.3	27.0	26.0	24.4	22.3	19.6
	10%	DB	21.7	24.7	26.8	30.3	32.5	33.2	33.1	32.8	31.7	29.2	26.3	22.3
		MCWB	18.6	20.7	22.4	25.0	26.2	27.1	27.2	26.9	25.8	24.1	21.7	18.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	23.1	25.0	27.6	28.4	29.0	29.2	28.7	28.1	26.6	25.1	22.6
		MCDB	24.6	27.3	29.2	32.2	34.1	33.7	33.3	33.1	32.1	30.1	28.1	25.7
	2%	WB	20.5	22.3	24.1	26.7	27.7	28.3	28.4	28.1	27.5	25.9	24.2	21.4
		MCDB	23.5	26.3	28.2	31.0	33.0	32.9	32.7	32.5	31.3	29.1	27.1	24.0
	5%	WB	19.7	21.7	23.5	26.1	27.3	27.9	27.9	27.7	27.0	25.4	23.5	20.3
		MCDB	22.4	25.4	27.2	30.2	32.2	32.4	32.3	31.8	30.7	28.5	26.2	23.1
	10%	WB	19.0	20.9	22.8	25.5	26.9	27.6	27.5	27.3	26.6	25.0	22.7	19.4
		MCDB	21.6	24.4	26.2	29.4	31.3	31.7	31.6	31.2	30.0	27.8	25.3	22.0
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.0	6.2	6.8	7.2	6.7	6.5	6.8	6.8	6.0	6.6	6.1
		MCDBR	8.0	9.5	8.6	9.0	9.4	8.7	8.5	8.7	8.4	7.9	8.4	8.1
		MCWBR	3.2	3.4	2.9	3.0	2.7	2.4	2.4	2.6	2.6	2.4	3.1	3.4
	5% WB	MCDBR	7.1	8.1	7.8	8.0	8.2	7.5	7.6	7.7	7.2	6.4	6.6	7.1
		MCWBR	3.3	3.4	3.0	3.0	2.8	2.6	2.7	2.8	2.5	2.4	3.0	3.5
(40) Clear-Sky Solar Irradiance	taub		0.576	0.620	0.851	0.801	0.547	0.507	0.500	0.563	0.590	0.659	0.580	0.613
	taud		1.780	1.717	1.347	1.472	2.053	2.187	2.195	1.994	1.895	1.696	1.818	1.703
	Ebn at Noon		681	685	557	602	772	797	802	755	726	645	673	628
	Edh at Noon		202	228	344	308	171	149	147	180	196	230	193	212
(44) All-Sky Solar Radiation	RadAvg		2.51	3.36	3.79	4.57	4.75	4.29	4.40	4.48	4.22	3.34	3.27	2.55
	RadStd		0.42	0.63	0.35	0.50	0.51	0.40	0.30	0.34	0.27	0.33	0.34	0.30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.35	N/A	-0.86	+0.63	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.40	N/A	N/A	+0.62	N/A	N/A	N/A	N/A	+193	+147

Nomenclature: See separate page